# CATERBOT - NOVEL AGRICULTURE TRACKED ROBOT FOR TRANSPORTATION OF COCONUTS - DESIGN AND ANALYSIS

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## Abstract

India is a country that has been primarily dependent on Agriculture as a livelihood for ages. Even today, more than 50% of the country's population is still dependent on Agriculture. Farmers, consisting most of this percentage, require unskilled labour for various agricultural activities. This unskilled labour adds additional economic burden to farmers, other than the already invested capital on pesticides, irrigation facilities, advanced machinery, etc. To counter this, we have developed a low-cost agriculture robot in order to assist farmers in an agricultural activities to relieve this financial pressure for them to a certain extent. Hence, we have developed a robot which commutes on caterpillar tracks(also known as continuous tracks) on the field and assist farmers. Solidworks was used to develop the robot under proper physical conditions. Its structural analysis and simulation under various constraints was carried out on ANSYS.

Keywords: Caterpillar tracks, Transportation, low-cost agriculture robot.

# 1 Introduction

Agriculture has been practiced in India for ages. Around 54.6% of India's population is dependent on agriculture for a living, thus making it the largest livelihood provider in India (IBEF - Agriculture and allied industries industry report, 2022). Agriculture also primarily affects India's economy, making it very important for the country's socioeconomic development (Green Earth Research Network, ). India leads in the production of milk, pulses, and jute, and ranks as the second largest producer of rice, wheat, sugarcane, groundnut, vegetables, fruit, and cotton. It is also one among the leading producers of spices, fish, poultry, livestock, and plantation crops (India at a glance, FAO in India, Food and Agriculture Organization of the United Nations, ). An estimated amount of 2,75,63,000 Hectares of land in India is allocated for 33,32,51,000 Megaton production of Horticulture Crops, estimated by Department of Agriculture and farmers welfare (Green Earth Research Network, ). Horticulture crops consists of fruits, vegetables, aromatics, medicinal, flowers, honey, spices, and plantation crops. After vegetables and fruits, plantation crops such as coconut and arecanut are the two which are produced the most in India when we talk about horticulture crops (Jithu et al., 2019). India's economy is largely dependent on the export of agricultural crops as well, as shown in Figure 1.

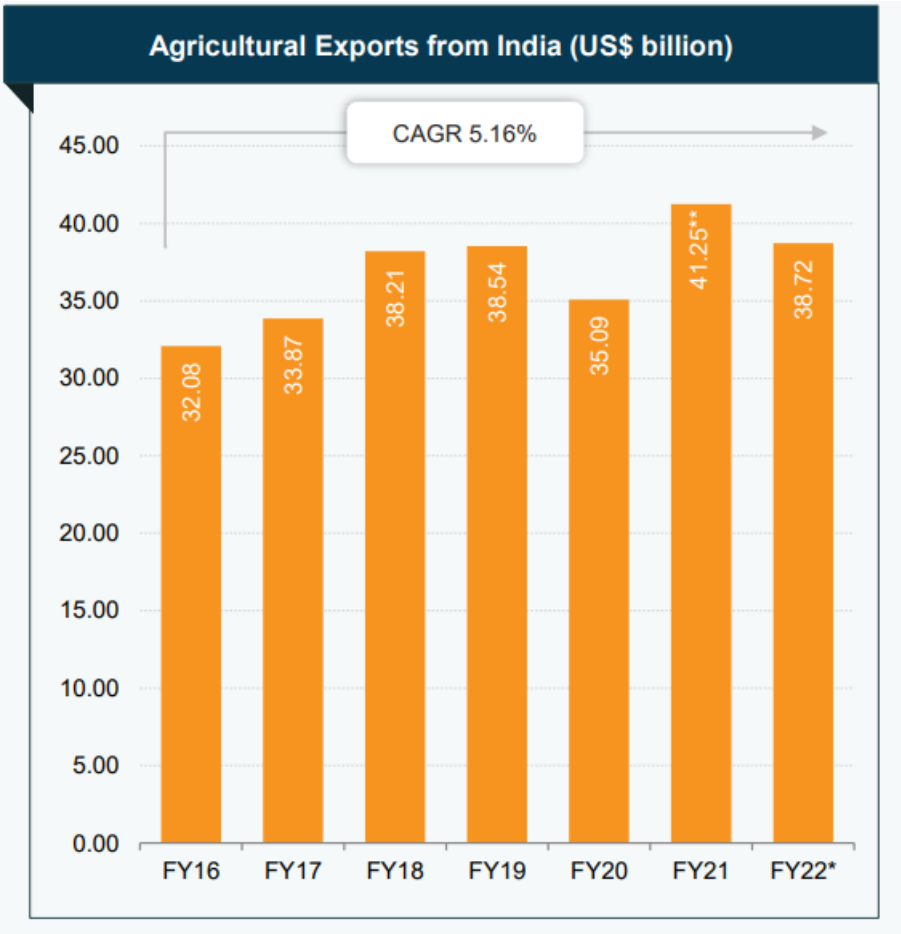



Figure 1: Agricultural Exports from India in US$ billion (IBEF - Agriculture and allied industries industry report, 2022)

Coconut is a very important plantation in India. The southern states of the country cultivate a lot of this plantation, and a significant amount of the population from the southern states is dependent on this crop

(Kappil et al., 2021). It is significantly cultivated in the coastal areas and on various other soils as well. (Srinivasan et al., 2021). Coconut also has numerous health benefits. Coconut oil has attracted attention due to its hypocholesterolemic, anticancer, antihepatosteatotic, antidiabetic, antioxidant, anti-inflammatory, antimicrobial and skin moisturizing properties (Deen et al., 2021). Coconut grows in large numbers in the state of Kerala, which is the leading state of Ayurveda in India. They also do find a lot of ayurvedic applications (Megalingam et al., 2017). In the past years, the number of coconuts produced have also grown a lot in number. Advancement in technology has lead to the development of unique harvest methods of coconuts (Jithu et al., 2019), (Sadeghi and Moradi, 2008). Not only for coconuts, but these unique harvesting methods have been deployed for other crops as well (Sebastian et al., 2015), (Aracil et al., 2003).

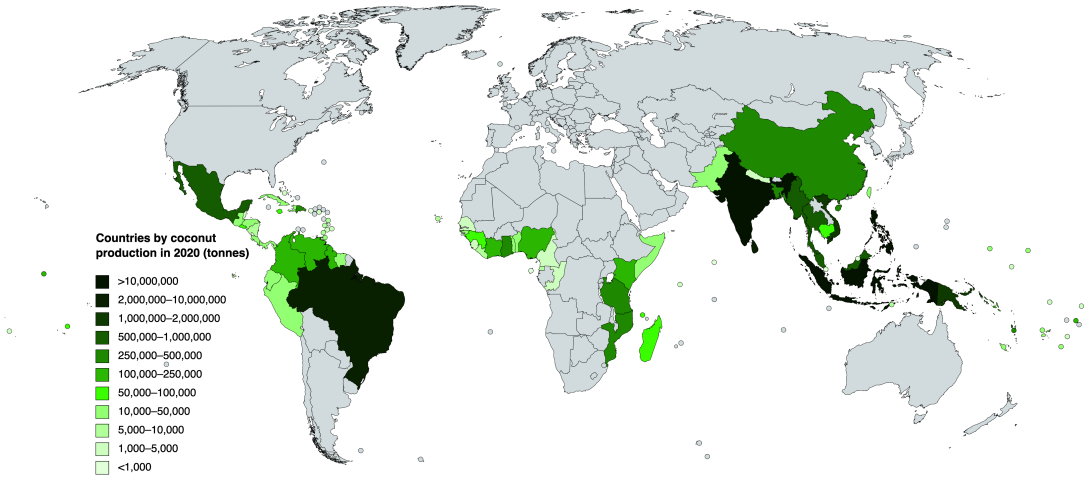



Figure 2: Countries by coconut production in 2020 (Source: Wikimedia Commons)

However, there has always been a traditional requirement for manual labour to harvest coconuts. The tall coconut trees have always demanded continuous and large amount of unskilled labour to be deployed for harvesting (Megalingam et al., 2017).

## 2 Inspiration for Caterbot

Carrying load in farms is mainly divided into two types: manual and using wheels. Manual labour is the most conventional way of carrying loads out in the fields. But in recent times, due to scarcity of labour, farmers are facing a lot of problems. The practice of manual load carrying is one of the dominant contributors to back pain and related problems (Kumar et al., 2021).

The other way of carrying loads is using wheels, in the sense of wagons, carts, tractors etc (Levin, 1960). Over 80% of farm produce is transported in animal-drawn carts (Yadav et al., 1998), but the challenges like human intervention and additional human effort, time-consuming process, cattle health deterioration, are making this option inefficient. Hence, usage of trucks, tractors, and other automobile vehicles with high mechanical efficiency is increasing day by day. Novel manufacturing techniques, navigation methods, tools and equipment, etc., have aided automobile companies to manufacture advanced agriculture machines which are suitable for almost all sorts of agricultural activities (Mocera and Somà, 2021). However, this requires additional investment and maintenance by farmers, plus a single machine might not be suitable for all agricultural activities, which will lead to farmers buying multiple machines, thus increasing economic burden. Also, these machines run on fossil fuels and the increasing use of fossil fuels is harmful to the environment (Hassan et al., 2021).

Coming to the current agriculture robots in use, the ones available in the market have been developed keeping in mind various use-cases (Bergerman et al., 2016). From harvesting to pesticide spraying to irrigation, robots have been developed for various agricultural activities (R Shamshiri et al., 2018). Recently, research breakthroughs in plant biology has made possible the artificial synthesis of new species of plant seeds, which can grow even in harsh climatic conditions and different soil profiles as well (Ravi and Anand, 2012). These advancements have agriculture spread to different topographical areas of the world. As a result, agricultural robots are also being deployed in different countries (Kushwaha et al., 2016). This has also lead to the development of robots for more specific agricultural tasks (Aravind et al., 2017).

However, a recent study shows the present terrestrial agriculture robots are mostly wheel-based (Oliveira et al., 2021). Wheel-based robots are not efficient because rubber tires may get stuck in a few soil profiles (Ding et al., 2011) Also, these robots generally navigate on differential drive mechanism (Maddahi et al., 2012). Differential drive mechanism does not provide the robot with all degrees of freedom at a single point of time. Hence, it would be more easier for the robot to get stuck in the soil, as the robot would require to take multiple turns and move in different directions. Also tire based vehicles cannot navigate on the uneven terrains in the fields in all cases effectively and might end up using a lot of power.

Keeping this main problem in mind, we have come up with a unique solution of devising a robot that uses caterpillar tracks instead of wheels. Caterpillar tracks are majorly seen being used in military tanks. Military tanks use caterpillar tracks so that they can operate in any terrain effectively. Caterpillar tracks have a continuous belt-like structure, which allows the vehicle using them to easily move onto uneven parts of the field (Polyakov et al., 2018). They have been already implemented in the past (Rowlinson, 1922) and are being implemented today as well (Teng, 2021). A few agricultural vehicles like tractors cab be spotted with caterpillar tracks (Ig and Melnikova, 2014). Caterpillar tracks are also being implemented for other transportation facilities (Dobretsov et al., 2021).

The robot has been named 'Caterbot'(Caterpillar track + Robot). Caterbot has been developed keeping three main objectives in mind:

1. It should be as light as possible.

2. It should be electric-driven.

3. It should be cost-effective such that farmers are able to afford it.

# 3 Caterbot - Design and Development

Caterbot has primarily been designed with the goal of transporting coconuts. Coconuts are heavy and sophisticated for transportation. Hence, we have chosen coconut and designed Caterbot in such a way that it would be helpful in carrying coconuts for farmers in an easier manner. It runs on a caterpillar track to avoid the difficulties a conventional rubber tire may cause as discussed above. It is fully electrical, lightweight, and cost-effective. Hence, most farmers can afford it. It has a rechargeable battery which helps it run for six hours on a single charge. The robot is suitable for any soil profile. It runs on four DC motors which power the gears. These gears interlock with the belt, thus avoiding slippage and increasing efficiency. Caterbot is suitable for operation in any climatic condition. It is corrosion-resistant.

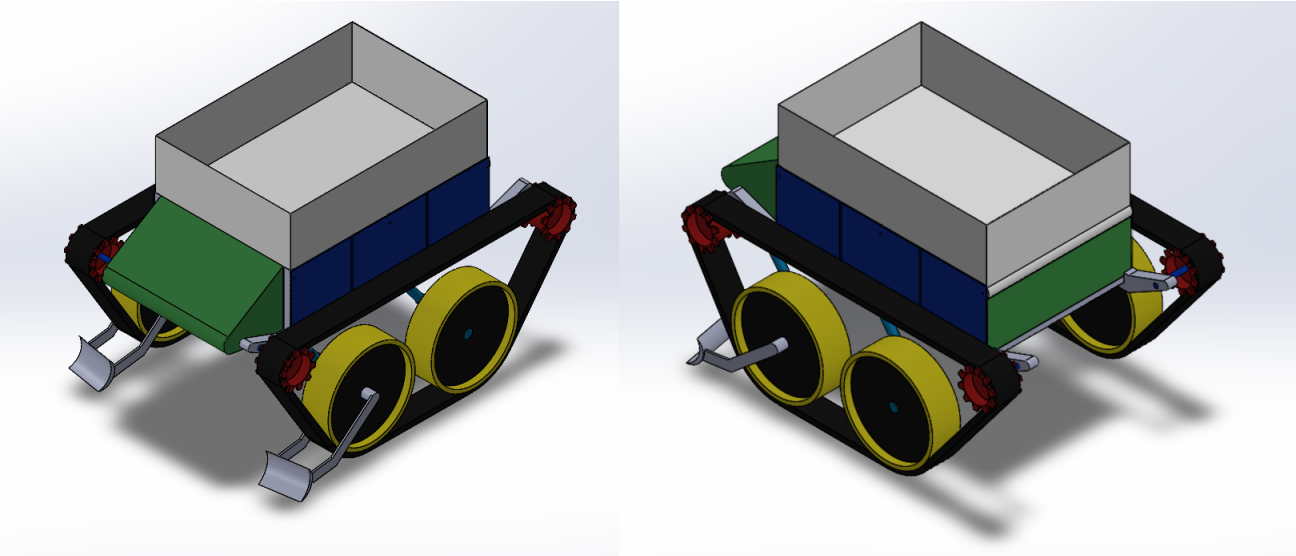


Figure 3: Front Isometric View of Caterbot(left) and Rear Isometric View of Caterbot(right)

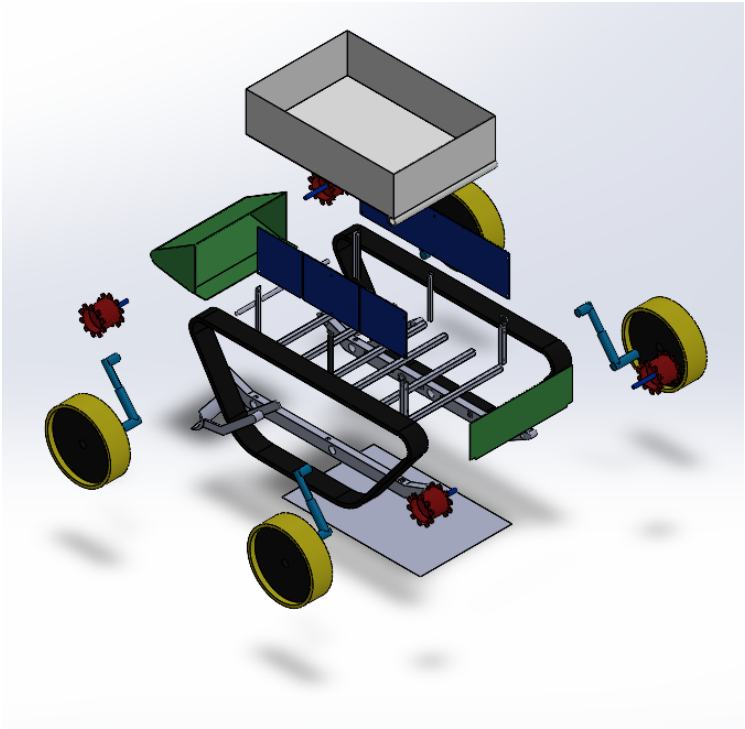
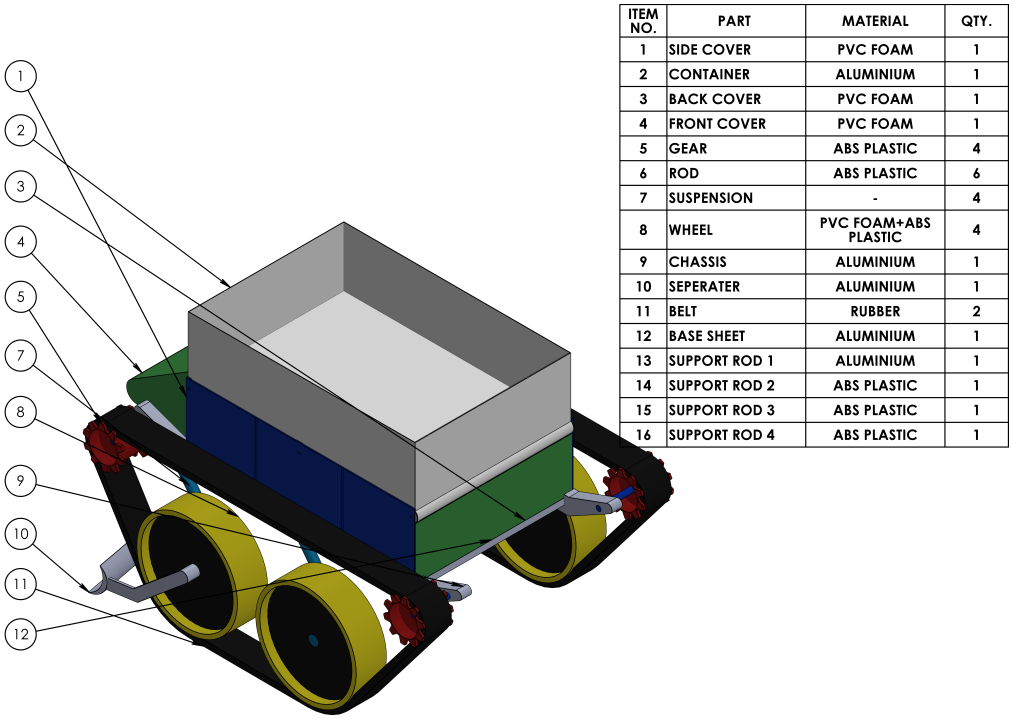


Figure 4: Exploded View of robot




| ITEM NO. | PART | MATERIAL | QTY. |
|---|---|---|---|
| 1 | SIDE COVER | PVC FOAM | 1 |
| 2 | CONTAINER | ALUMINIUM | 1 |
| 3 | BACK COVER | PVC FOAM | 1 |
| 4 | FRONT COVER | PVC FOAM | 1 |
| 5 | GEAR | ABS PLASTIC | 4 |
| 6 | ROD | ABS PLASTIC | 6 |
| 7 | SUSPENSION | - | 4 |
| 8 | WHEEL | PVC FOAM+ABS PLASTIC | 4 |
| 9 | CHASSIS | ALUMINIUM | 1 |
| 10 | SEPERATER | ALUMINIUM | 1 |
| 11 | BELT | RUBBER | 2 |
| 12 | BASE SHEET | ALUMINIUM | 1 |
| 13 | SUPPORT ROD 1 | ALUMINIUM | 1 |
| 14 | SUPPORT ROD 2 | ABS PLASTIC | 1 |
| 15 | SUPPORT ROD 3 | ABS PLASTIC | 1 |
| 16 | SUPPORT ROD 4 | ABS PLASTIC | 1 |

Figure 5: Drawing Sheet of Caterbot

Table 1: Mechanical properties of parts of Caterbot.

| S No. | Part | Material | Quantity | Mass(per unit in kg) |
|---|---|---|---|---|
| 1 | Side Cover | PVC Foam | 1 | 0.13958 |
| 2 | Container | Aluminium | 1 | 14.121 |
| 3 | Back Cover | PVC Foam | 1 | 0.080016 |
| 4 | Front Cover | PVC Foam | 1 | 0.30746 |
| 5 | Gear | ABS Plastic | 4 | 1.39924 |
| 6 | Rod | ABS Plastic | 6 | 0.16751 |
| 7 | Suspension | - | 4 | 2 |
| 8 | Wheel | PVC Foam + ABS Plastic | 4 | 5.64453 |
| 9 | Chassis | Aluminium | 1 | 15.615 |
| 10 | Separator | Aluminium | 1 | 2.925 |
| 11 | Belt | Synthetic Rubber | 2 | 15 |
| 12 | Base Sheet | Aluminium | 1 | 2.0043 |
| 13 | Support Rod 1 | Aluminium | 1 | 0.15 |
| 14 | Support Rod 2 | ABS Plastic | 1 | 0.058084 |
| 15 | Support Rod 3 | ABS Plastic | 1 | 0.047075 |
| 16 | Support Rod 4 | ABS Plastic | 1 | 0.094005 |
| 17 | Battery | Lithium Ion | 1 | 25 |
| 18 | Motors | DC Motors | 4 | 0.5 |
| TOTAL | | | | 129.72166 |

Total Weight of the robot at no load condition = 129.72166 Kgs
Parts like the Container and Chassis are made of aluminium since aluminium is lightweight and can withstand good amount of stress. ABS plastic, PVC foam and synthetic rubber has also been chosen with the same

idea.

### 3.1 Capacity of Caterbot

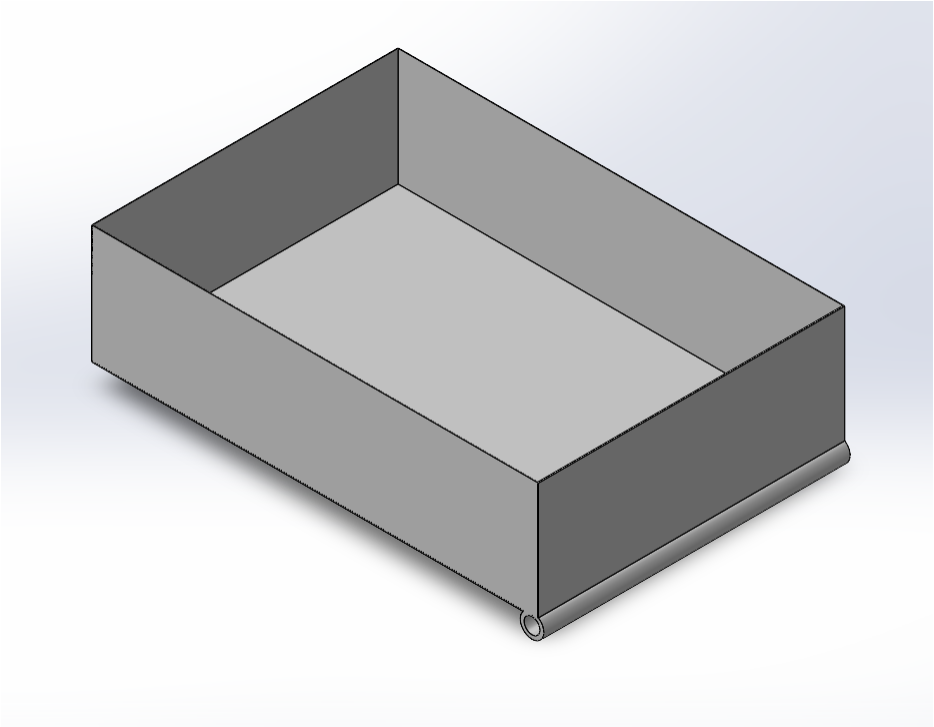


Figure 6: Container of Caterbot

Bounding Box of Container: $1.126 * 0.773 * 0.3$ (All dimensions are in meters)

Avg dimensions of coconut= $0.2(length) * 0.145(diameter)$ (Amudhayazhini, )

No.of coconuts that can fit in the container along its length=$\frac{1.126}{0.2} = 5$

No.of coconuts that can fit in the container along its width=$\frac{0.773}{0.145}$=5

No.of Coconuts in 1st stack = 25

No.of Coconuts in 2nd stack = 16

No.of Coconuts in 3rd stack = 9

Total coconuts in the container = 50

Avg. weight of green coconut = 1 kg

Max weight of green coconut = 1.5 kg

Avg payload carried by the robot in one cycle = 50 kg

Max payload the robot can carry in one cycle = 75 kg

Therefore, maximum payload can be taken as 80 kg.

### 3.2 Motor Parameters

#### 3.2.1 Torque

$$T = F * r$$

$$Where: T = Torque$$

$$F = Force\ with\ which\ the\ wheel\ rotates$$

$$r = Radius\ of\ the\ wheel$$

$$T = m * a * r$$

$$Where: m = Mass\ of\ the\ robot$$

$$a = Acceleration\ of\ the\ robot$$

$$r = Radius\ of\ \ the\ wheel$$

$$T = 220 * 0.3 * 0.075$$

$$T = 5\ Nm$$

The torque calculated above will not be available completely for functioning Caterbot, as a certain part of it will go into frictional losses in the motor. The torque which is available for doing useful work is known as shaft torque $T_{sh}$. It is so called because it is available at the shaft. Most electric motors are designed to run at 50% to 100% of the rated load. Maximum efficiency is usually near 75% of the rated load.

Therefore,

$$0.75 * T_{motor} = T_{wheel}$$

$$T_{motor} = \frac{T_{wheel}}{0.75}$$

$$T_{motor} = \frac{5}{0.75}$$

$$T_{motor} = 7\ Nm$$

#### 3.2.2 Angular Velocity

$$w(Angular\ velocity) = \frac{v}{r}$$

$$Where: v = velocity\ with\ which\ the\ robot\ moves$$

$$r = radius\ of\ the\ wheel$$

$$w = \frac{v}{r}$$

$$w = \frac{0.3}{0.075}$$

$$w_{motor} = 4\ \frac{rad}{s} = 38.2\ rpm$$

.

### 3.2.3 Power Consumed

$$Power\ consumed = T * w$$

$$Where: T = Torque$$

$$w = Angular\ velocity\ of\ motor$$

$$P = T * w$$

$$P = 7 * 2 * 3.14 * \frac{40}{60} = 30\ W$$

| Motor Parametres | |
|---|---|
| Torque | 7 Nm |
| Angular Velocity | 40 rpm |
| Power Consumed | 30 W |

## 3.3 Battery Specifications

$$Operating\ Voltage = 12\ V$$

$$Battery\ capacity\ required = I * t$$

$$Where: I = current$$

$$t = operating\ time\ in\ hrs$$

$$Considering\ 50\%\ battery\ efficiency$$

$$= \frac{30}{12} * \frac{6}{0.5}$$

$$Required\ battery\ per\ motor = 30\ Ah$$

| Battery Specifications | | |
|---|---|---|
| 1 | Capacity | 120Ah |
| 2 | Operating Voltage | 12 V |
| 3 | Battery Cell Composition | Lithium Ion |
| 4 | Item Weight | 25 Kg |

# 4 Design Simulation and Analysis

Static Structural Analysis was carried out on Caterbot to examine whether the robot is able to accomplish the necessary goals. The following static structural conditions were imposed on the robot as shown in Figure 7.

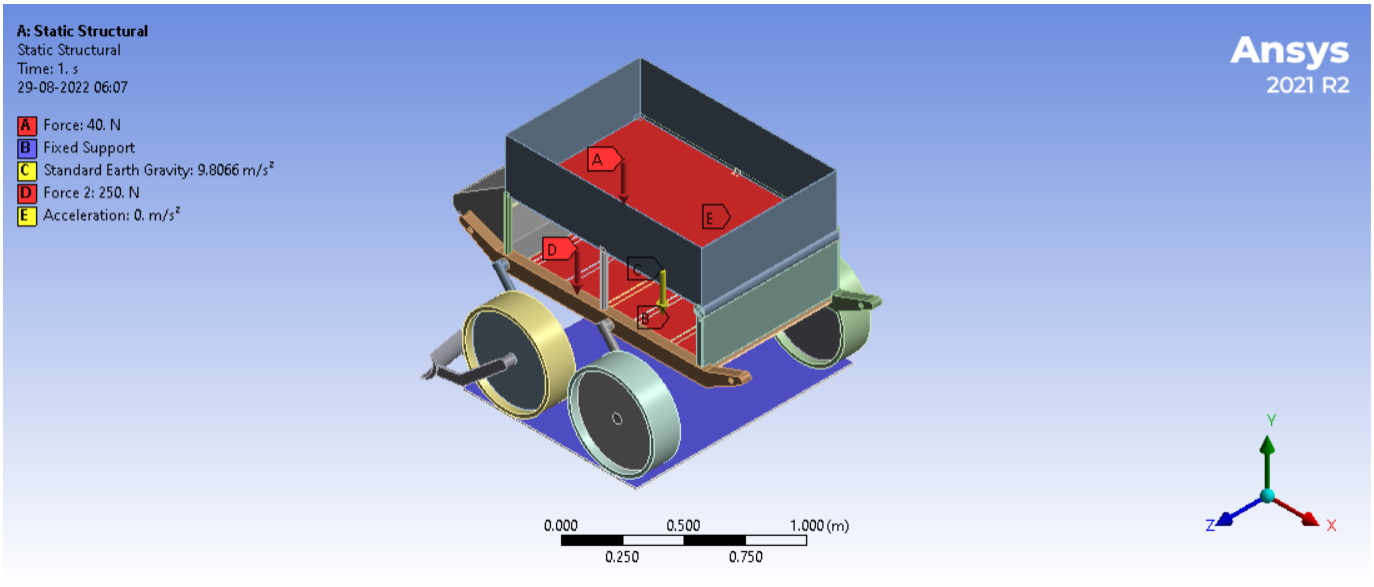



Figure 7: Static Structural Analysis of Caterbot in ANSYS

Table 2: Static structural analysis of Caterbot

| Parameter | Static Structural Conditions | Explanation | Value |
|---|---|---|---|
| A | Force | Dynamically increasing payload weight acting on the container | 40 to 800 N |
| B | Fixed Support | The base is acting as a field | Fixed |
| C | Earth Gravity | Standard earth gravity acting on the Caterbot | $9.8\frac{m}{s^2}$ |
| D | Force | The weight of the battery and other electrical components acting on the chassis | 250 N |
| E | Acceleration of the Caterbot | The acceleration with which Caterbot moves in the wet fields | $0\ to\ 0.3\frac{m}{s^2}$ |

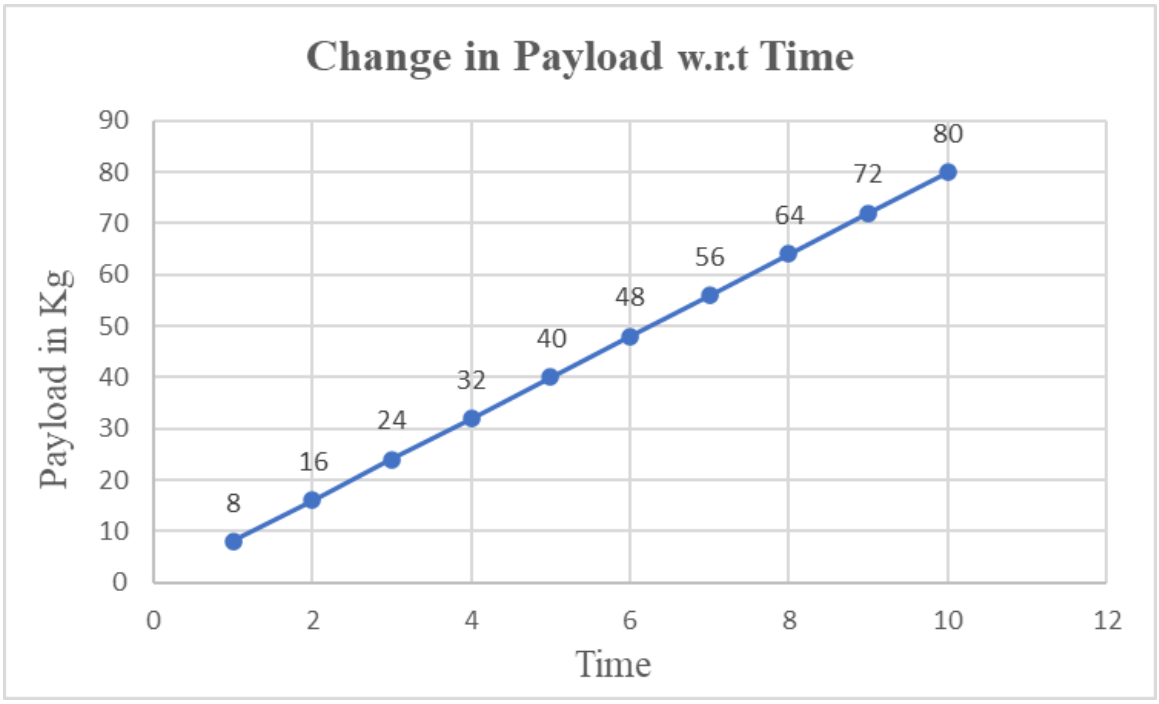



Figure 8: Change in Payload w.r.t Time

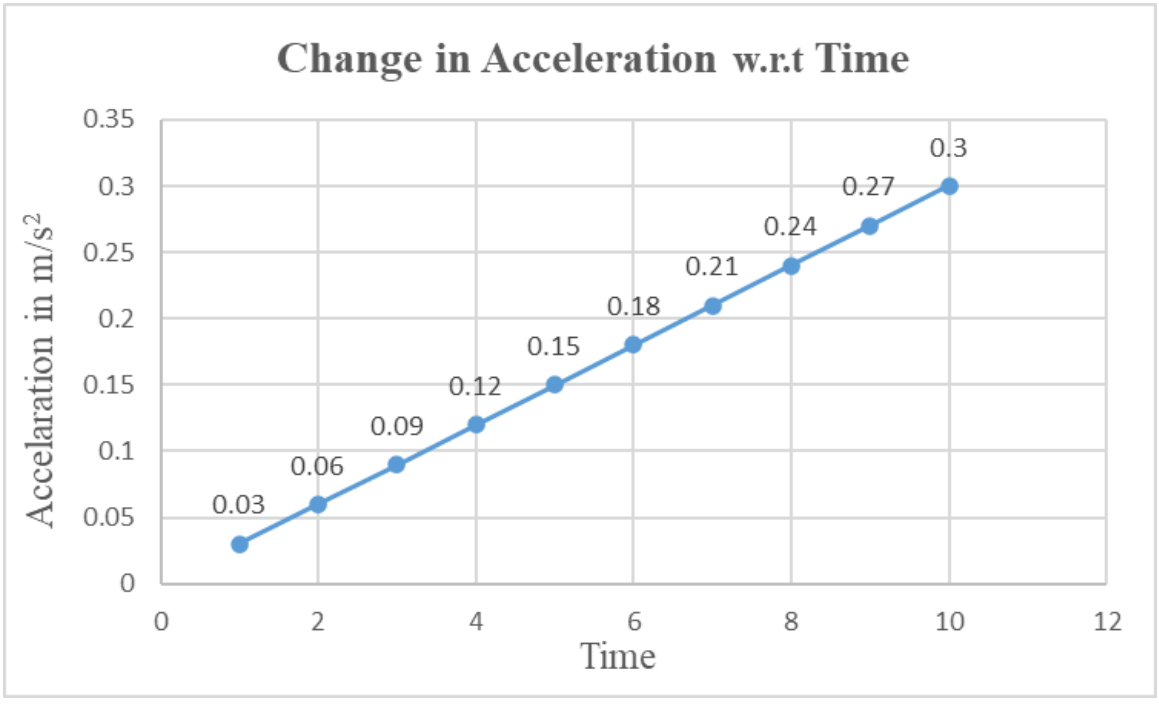



Figure 9: Change in Acceleration w.r.t Time

Table 3: Mechanical properties of materials used in Caterbot

| Mechanical Property | Material | |
|---|---|---|
| | Aluminium 6061 | ABS Plastic |
| Yield Strength in MPa | 276 | 15-65 |
| Elongation at break | 15% | 3% |
| Allowable Elastic Strain | 0.05 | 0.01 |

# 5 Result and Discussions

After the simulation, the following results were obtained- Total deformation, directional deformation, equivalent stress and equivalent elastic strain were the parameters that were analysed while carrying out the static structural analysis of Caterbot with a payload of 80 Kgs and the results are shown in figures.

## 5.1 Total Deformation

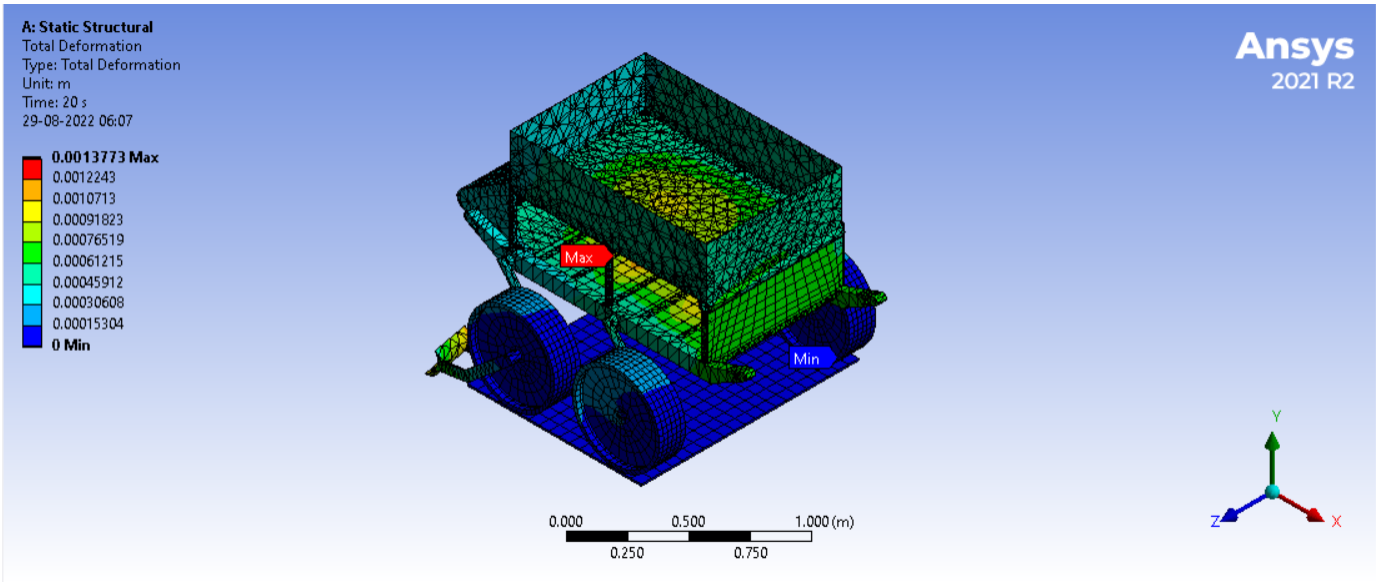



Figure 10: Total deformation of Caterbot in ANSYS

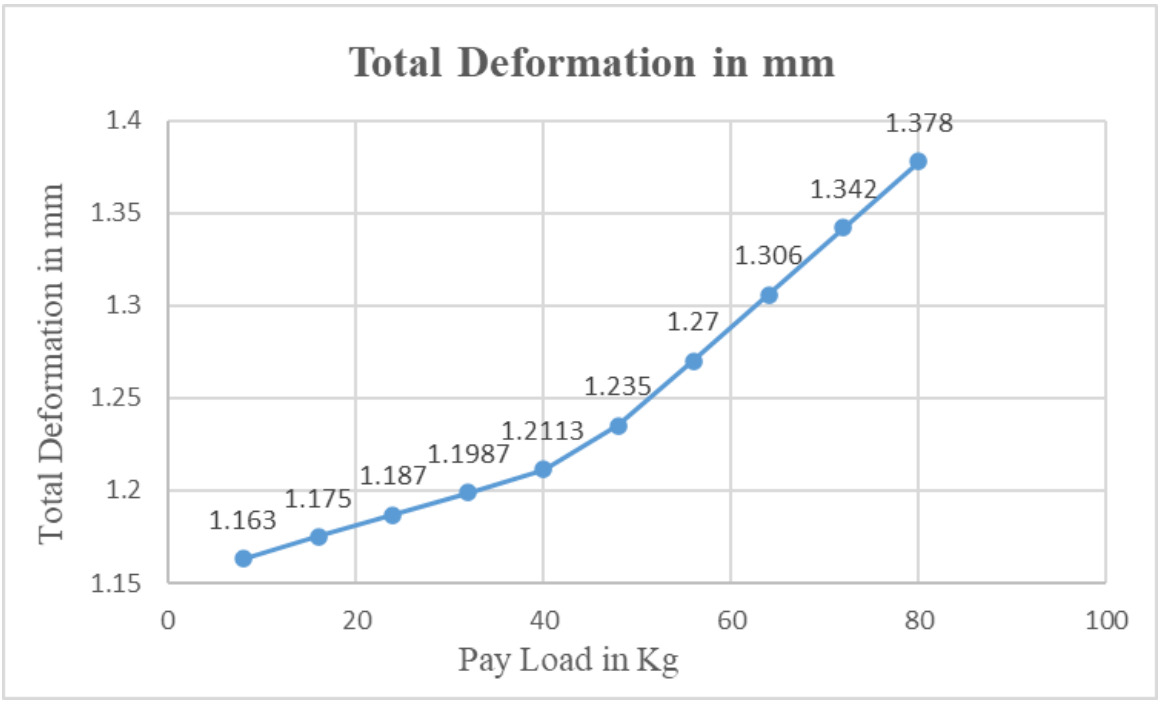



Figure 11: Total deformation of the Caterbot (The labels show the value of total deformation corresponding to the payload)

Figure 11 shows the relationship between the value of total deformation on Caterbot and the applied payload. It can easily be observed that total deformation is approximately directly proportional to the applied payload. The value of total deformation increases proportionally with the applied payload.

Table 4: Total deformation for respective payloads

| S No. | Applied Payload (in Kg) | Deformation in mm |
|---|---|---|
| 1 | 8 | 1.163 |
| 2 | 16 | 1.175 |
| 3 | 24 | 1.187 |
| 4 | 32 | 1.1987 |
| 5 | 40 | 1.2113 |
| 6 | 48 | 1.235 |
| 7 | 56 | 1.27 |
| 8 | 64 | 1.306 |
| 9 | 72 | 1.342 |
| 10 | 80 | 1.378 |

Table 4 shows the value of total deformation at different applied payloads. When a payload of 80 kg was applied, the total deformation was found to be 1.378 mm which is the permissible deformation, as shown in Table 3.

### 5.2 Directional Deformation

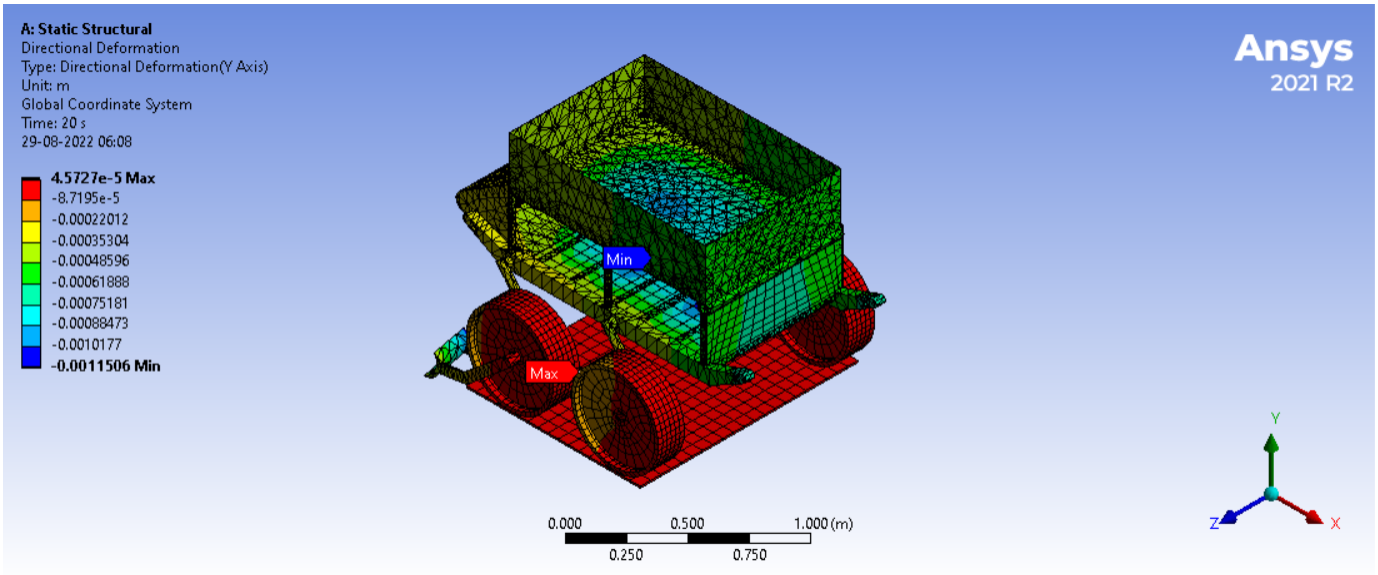



Figure 12: Directional deformation of Caterbot in ANSYS

As shown in Figure 12 the maximum value of the directional deformation along the Y-Axis is 1.1506mm, Which is at the centre of the container of Caterbot under the applied loading conditions.

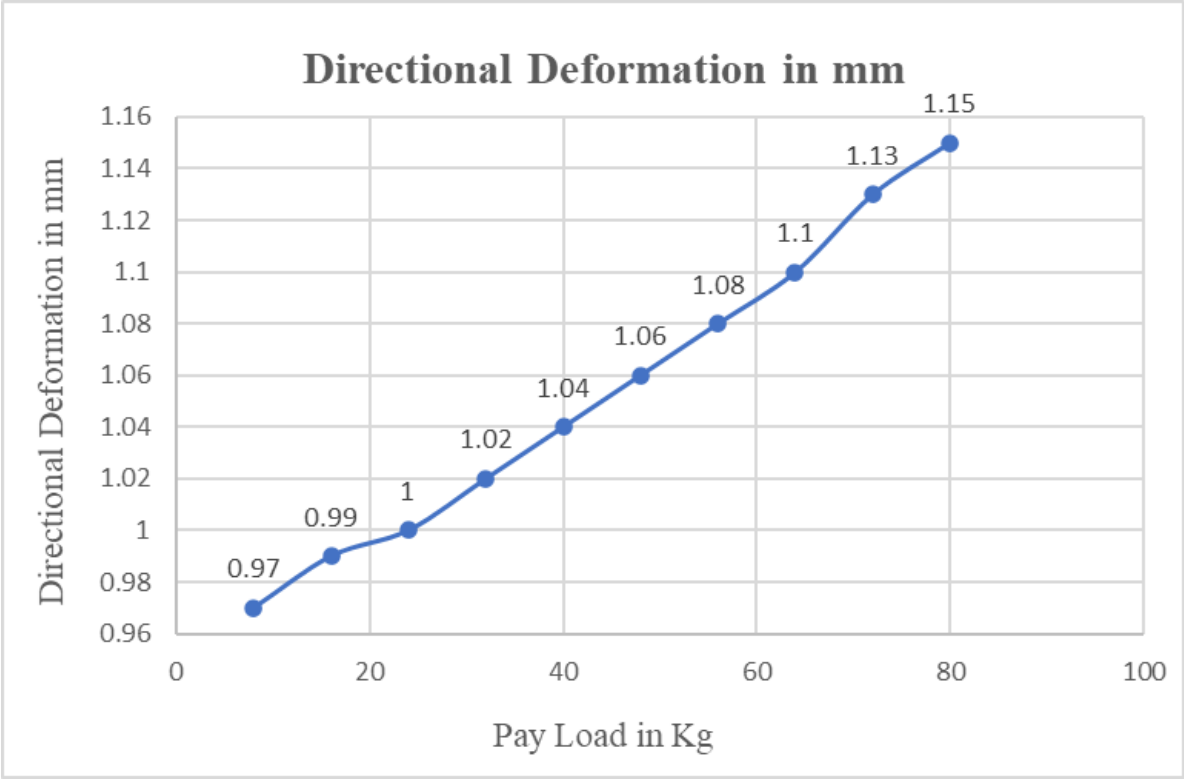



Figure 13: Directional deformation for the Caterbot (The labels show the value of directional deformation corresponding to the payload)

Figure 13 shows the relationship between the value of directional deformation on the robot and the applied payload. It can easily be observed that directional deformation is approximately directly proportional to the applied payload. The value of directional deformation increases proportionally with the applied payload.

Table 5: Directional deformation for respective payloads

| S No. | Applied Payload (in Kg) | Deformation in mm |
|---|---|---|
| 1 | 8 | 0.97 |
| 2 | 16 | 0.99 |
| 3 | 24 | 1 |
| 4 | 32 | 1.02 |
| 5 | 40 | 1.04 |
| 6 | 48 | 1.06 |
| 7 | 56 | 1.08 |
| 8 | 64 | 1.1 |
| 9 | 72 | 1.13 |
| 10 | 80 | 1.15 |

Table 5 shows the value of directional deformation at different applied payloads. When a payload of 80 kg was applied, the directional deformation was found to be 1.15 mm which is the permissible deformation, as shown in Table 3.

### 5.3 Equivalent Stress

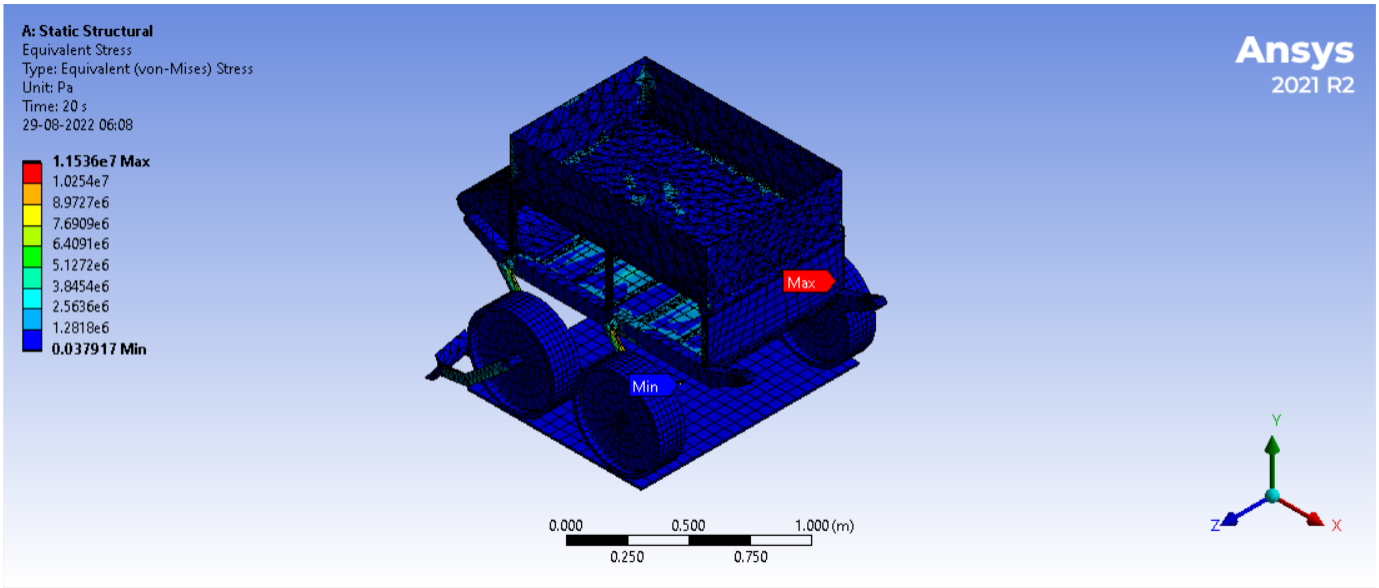



Figure 14: Stress analysis of Caterbot in ANSYS

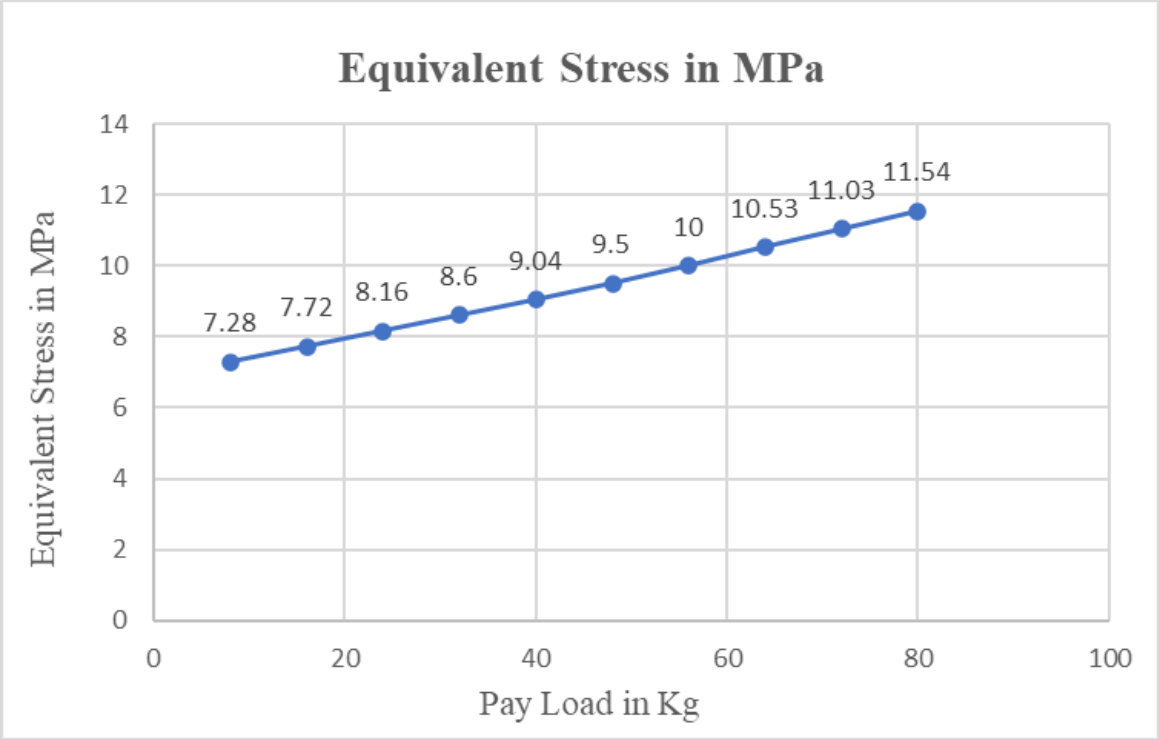



Figure 15: Equivalent Stress for Caterbot (The labels show the value of maximum equivalent stress corresponding to the payload)

Figure 15 shows the relationship between the value of equivalent stress on Caterbot and the applied payload. It can easily be observed that equivalent stress is directly proportional to the applied payload i.e., the value of maximum equivalent stress increases proportionally with the applied payload.

Table 6: Equivalent stress for respective payloads

| S No. | Applied Payload (in Kg) | Equivalent Stress in MPa |
|---|---|---|
| 1 | 8 | 7.28 |
| 2 | 16 | 7.72 |
| 3 | 24 | 8.16 |
| 4 | 32 | 8.6 |
| 5 | 40 | 9.04 |
| 6 | 48 | 9.5 |
| 7 | 56 | 10 |
| 8 | 64 | 10.53 |
| 9 | 72 | 11.03 |
| 10 | 80 | 11.54 |

Table 6 shows the value of equivalent stress at different applied payloads. When a payload of 80 kg was applied, the maximum equivalent stress was found to be 11.54 MPa, which is the permissible stress value as shown in Table 3.

### 5.4 Equivalent Elastic Strain

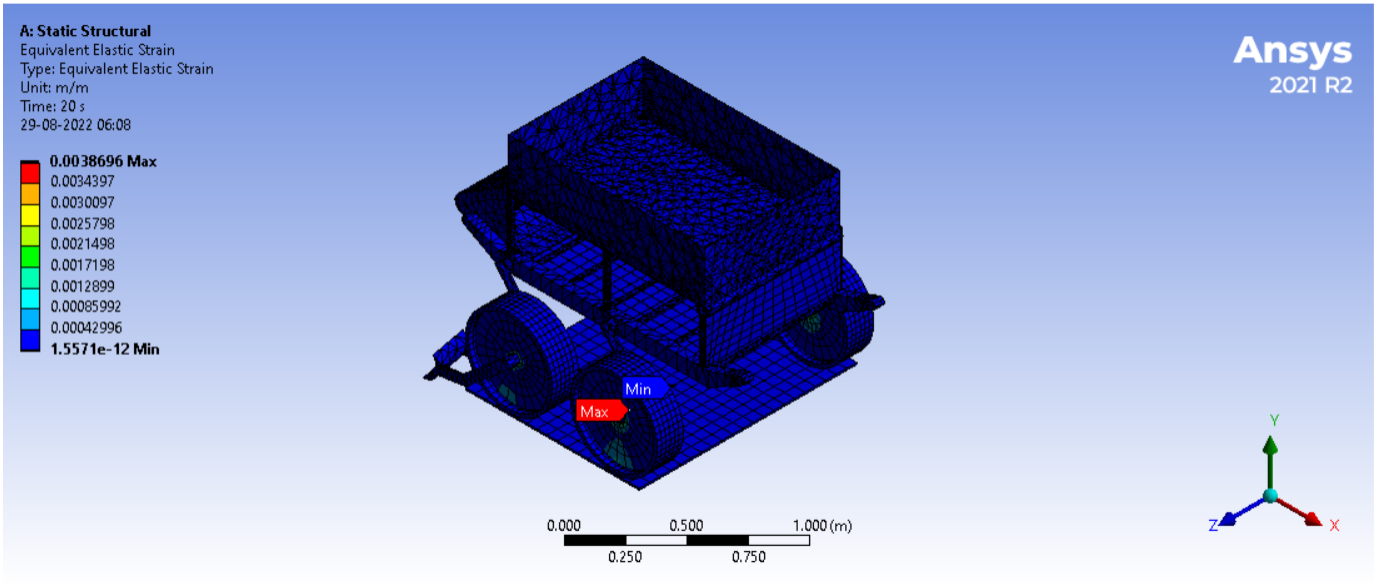



Figure 16: Strain analysis of Caterbot in ANSYS

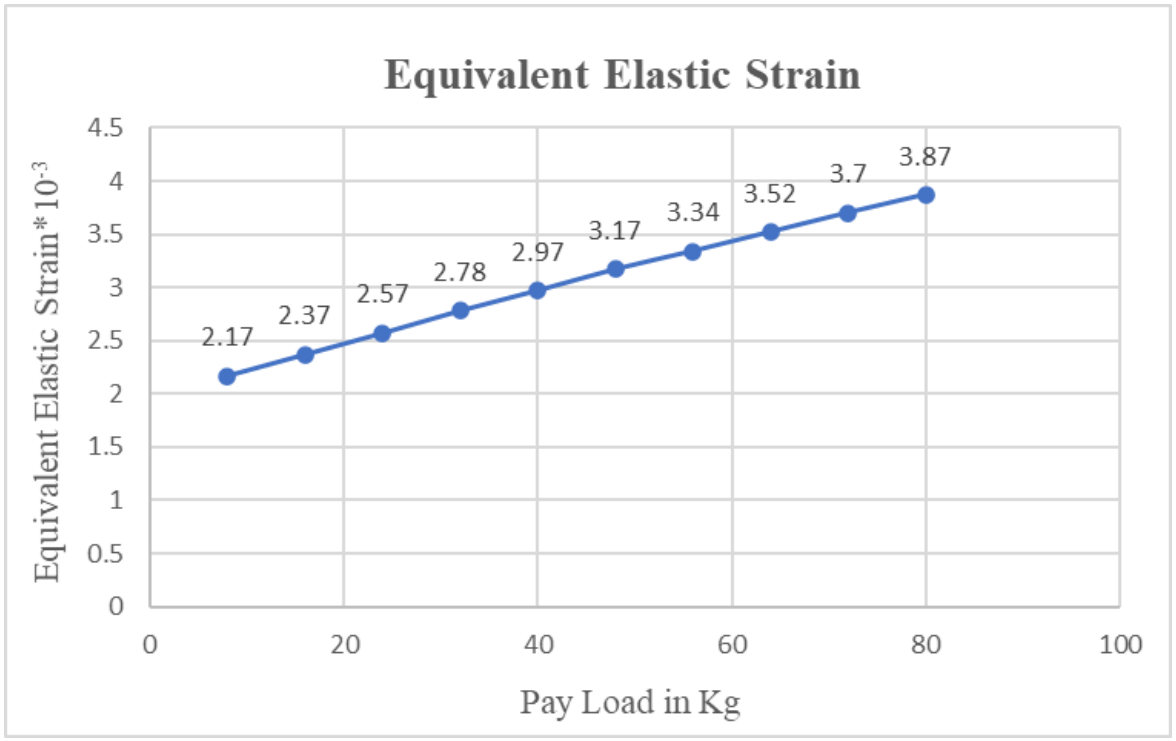



Figure 17: Equivalent elastic strain for Caterbot (The labels show the value of equivalent elastic strain corresponding to the payload)

Figure 17 shows the relationship between equivalent elastic strain on Caterbot and the applied payload. It can easily be observed that equivalent elastic strain is directly proportional to the applied payload i.e., the

value of equivalent elastic strain increases proportionally with the applied payload is increased.

Table 7: Equivalent strain for respective payloads

| S No. | Applied Payload (in Kg) | $Equivalent\ Strain * 10^{-3}$ |
|---|---|---|
| 1 | 8 | 2.17 |
| 2 | 16 | 2.37 |
| 3 | 24 | 2.57 |
| 4 | 32 | 2.78 |
| 5 | 40 | 2.97 |
| 6 | 48 | 3.17 |
| 7 | 56 | 3.34 |
| 8 | 64 | 3.52 |
| 9 | 72 | 3.7 |
| 10 | 80 | 3.87 |

Table 7 shows the value of equivalent elastic strain at different applied payloads. When a payload of 80 kg was applied, the equivalent elastic strain was found to be 3.87, which is the permissible strain value as shown in Table 3.

### 5.5 Overall Change in Parameters

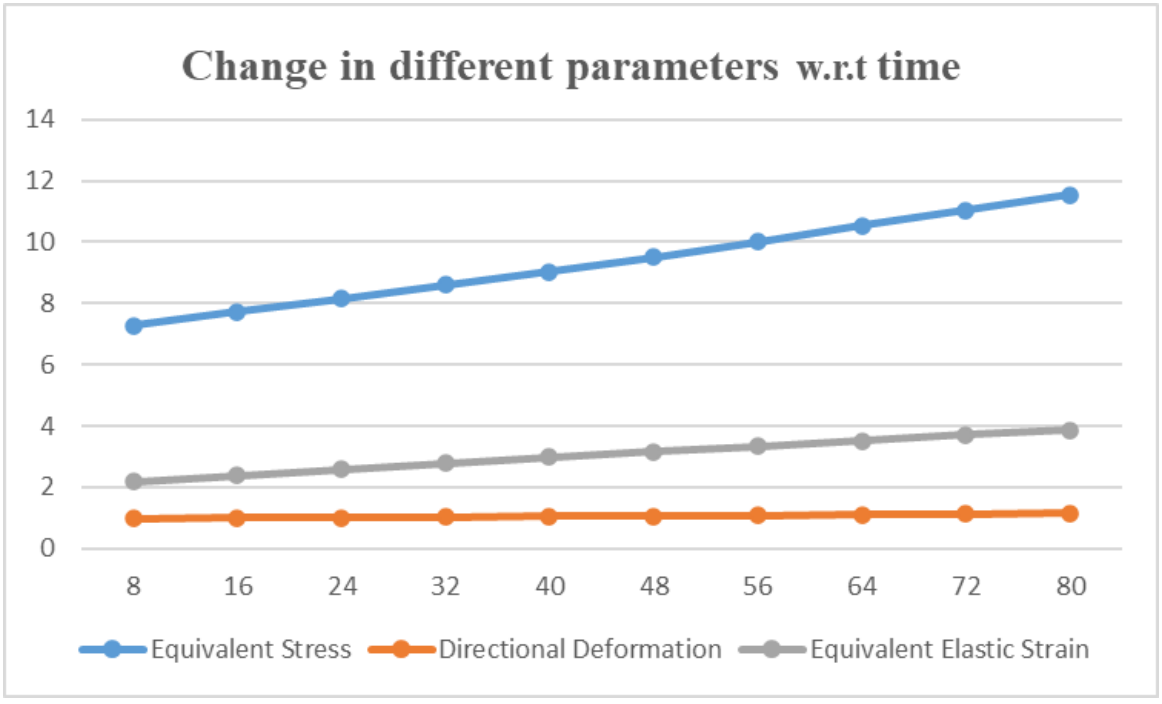



Figure 18: Change in different Parameters w.r.t payload

Table 8: Change in different parameters w.r.t payload

| S No. | Payload (in Kg) | Directional Deformation (in mm) | Equivalent Stress (in MPa) | Equivalent Elastic Strain |
|---|---|---|---|---|
| 1 | 8 | 0.97 | 7.28 | 2.17 |
| 2 | 16 | 0.99 | 7.72 | 2.37 |
| 3 | 24 | 1 | 8.16 | 2.57 |
| 4 | 32 | 1.02 | 8.6 | 2.78 |
| 5 | 40 | 1.04 | 9.04 | 2.97 |
| 6 | 48 | 1.06 | 9.5 | 3.17 |
| 7 | 56 | 1.08 | 10 | 3.34 |
| 8 | 64 | 1.1 | 10.53 | 3.52 |
| 9 | 72 | 1.13 | 11.03 | 3.7 |
| 10 | 80 | 1.15 | 11.54 | 3.87 |

Figure 18 shows the results in a single graph, making it easy to infer for as to how Caterbot reacted to the applied conditions. The graph shows a clear variation of the three analysed parameters with the variation of the applied payload.

# 6 Conclusion

The above results show the maximum permissible strain, maximum permissible stress, and maximum permissible deformation. Looking at the values, it can be concluded Caterbot is safe under the following conditions, as discussed in Table 2. Hence, Caterbot will be successful in transporting the estimated number of coconuts(according to Caterbot's container's volume) within the field. This concludes that the objectives of developing the robot with features such as lightweight, electrically driven and economically feasible - will be achieved if Caterbot is deployed in the following manner on the coconut field.

# 7 Future Scope

However, Project Caterbot has long-term goals. It has been started with the primary goal of making an economically feasible assistant robot to farmers, especially the small and marginal ones, as discussed above. The future versions of the project will mainly aim at assisting farmers not only for the transport of coconuts, but also other crops as well. The further versions of the robot will also be developed in such a way that they can be used for other agricultural activities as well. They will come up with features like a hydraulic mechanism for unloading the crop yield, proximity sensors for obstacle detection(for better navigation) and identifying various objects, better builds to carry more yield, spray guns for supplying pesticides to the crops, plough for soil tilling. The project also aims at making the next robots customisable according to the farmer's needs. Ultimately, the project envisions to make a robust robot which is a replacement for existing sophisticated and high-cost agricultural machinery.

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